Guetteur: active perception with receipts

Team Drift. OpenCV AI Competition 2026, powered by AWS. Agentic Vision path. Technical report, updated 2026-10-15.

> **Status of the evidence.** Accuracy and comparative benchmark figures are from **synthetic** depth scenes on an M4 Pro. The AWS judge endpoint is deployed and fresh synthetic smoke results are included below. Physical OAK-D capture has been verified for three 300-frame takes; calibrated replay and real-world accuracy are **not measured**. The public AWS demo does not ingest live camera data. The dashboard's gate selector simulates approval decisions, not an authenticated cloud approval workflow.

1. Summary

Guetteur ("lookout" in French) is a privacy-first agent that watches a space through **depth only**. It uses **OpenCV 5** to turn depth frames into small, picture-free evidence, lets an agent choose what to do *next* from that evidence, stops at a **human gate** before anything outward-facing happens (silence means deny), and attaches a **re-runnable receipt** to every claim it makes, in the prove-it-done format.

What the evaluation shows (synthetic, 11 scenarios x 3 seeds = 33 runs per planner):

	Guetteur agent	Threshold-and-alert baseline
Scenarios succeeded	**30/33** (every detectable case; the 3 misses are one documented sub-noise scenario)	12/33
Event precision	**1.00** (0 false alerts)	0.25 (27 false alerts)
Event recall	**0.86** (1.00 excluding the documented miss)	0.43
Honest receipt ledgers proven	**21/21**	n/a
Doctored claims rejected	**24/24**	n/a

The first contribution is the closed loop: an ambiguous observation changes *how OpenCV is called next* (frames integrated, threshold), and the same evidence sizes the request to a human. The second is that no claim is accepted on the agent's word: two independent estimators re-measure it against a pinned input and a doctored claim fails.

2. Problem

Camera-based AI usually means shipping pictures to someone else's cloud and trusting the system's summary. Two things go wrong. People's images leave the room, and agents report things nobody can

check ("zone clear", "item removed"), including when the sensor itself is broken, drifting, or looking at a person it mistakes for a change.

Intended users: small operators who want to know "did something change on that shelf, in that equipment room, at that door?" without keeping a video archive (stockrooms, labs, makerspaces, network closets, community tool libraries); homes and care settings where occupancy and change matter but pictures are unacceptable (opt-in, with consent); and builders of agent systems who need a reference for *verifiable* vision claims.

We make no statistical claim about harm reduced. What we demonstrate and measure: a vision agent that works from depth only (there is no image to leak), fewer false alarms and silent failures than a threshold-and-alert pipeline on the same evidence, and claims that anyone can re-check.

3. Architecture

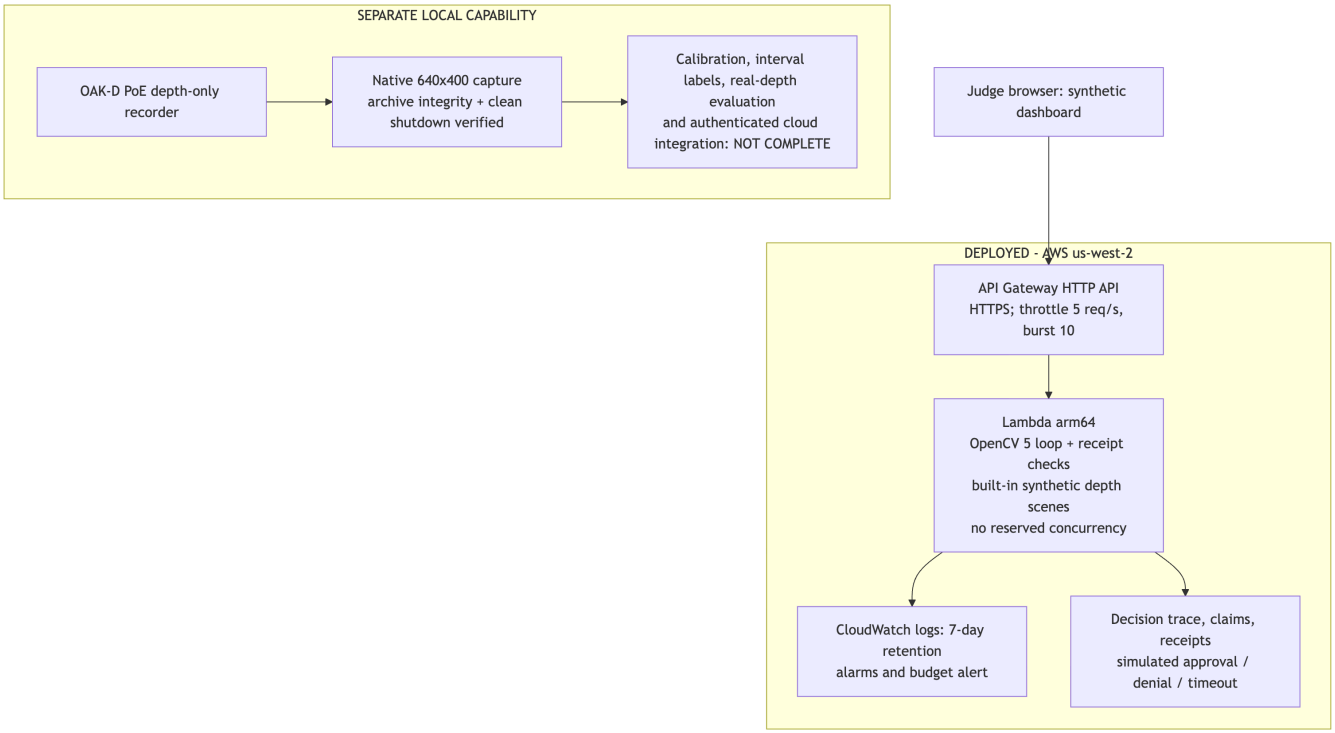


Figure 1. Deployed AWS judge path and separate physical recorder. The public demo runs built-in synthetic scenes. There is no deployed path from the physical recorder to the cloud detector or an authenticated approval service. The perception-decision-action loop is detailed in Figure 2.

Part	Status
Perception, planner, executor, human gate, claims, receipts (Python 3.12, OpenCV 5.0.0)	built; synthetic evaluation and receipt/tamper tests recorded in the repository
Judge dashboard + API (`/`, `/healthz`, `/api/scenarios`, `/api/run`, `/api/verify`) as one Lambda handler, also served locally	built; tested locally and on a linux/arm64 Lambda base image
AWS: API Gateway HTTP API + Lambda arm64 + CloudWatch + budget alert, one SAM stack	deployed; fresh public `/healthz` reports OpenCV 5.0.0 on aarch64; smoke evidence in `docs/submission/`
Authenticated evidence intake, approvals, DynamoDB with TTL, private S3 for opt-in depth snippets	phase 2: code and local tests only, deliberately **not** in the first deploy
Real OAK-D depth input and clip recorder	opt-in native recorder verified on hardware; three 300-frame takes archived locally; calibrated replay remains open

Privacy by construction. The camera's colour sensor feeds only the on-chip detector; only stereo depth reaches the host. No code path reads, writes or sends a picture, and a test scans code, data and run outputs for image/video files and imaging calls. Depth still reveals occupancy, silhouettes at close range and timing, so event data is treated as sensitive. Evidence that would leave the device is JSON (scalars, up to three blobs, a 6x8 signed-millimetre grid, at most 16 KB) and a schema rejects anything pictorial.

What the public endpoint serves. Built-in synthetic scenes only, labelled `synthetic: true` in the page and in every response. It accepts scenario names and small JSON; no uploads, no visitor-chosen paths, body size limited.

4. The agentic loop and prove-it-done receipts

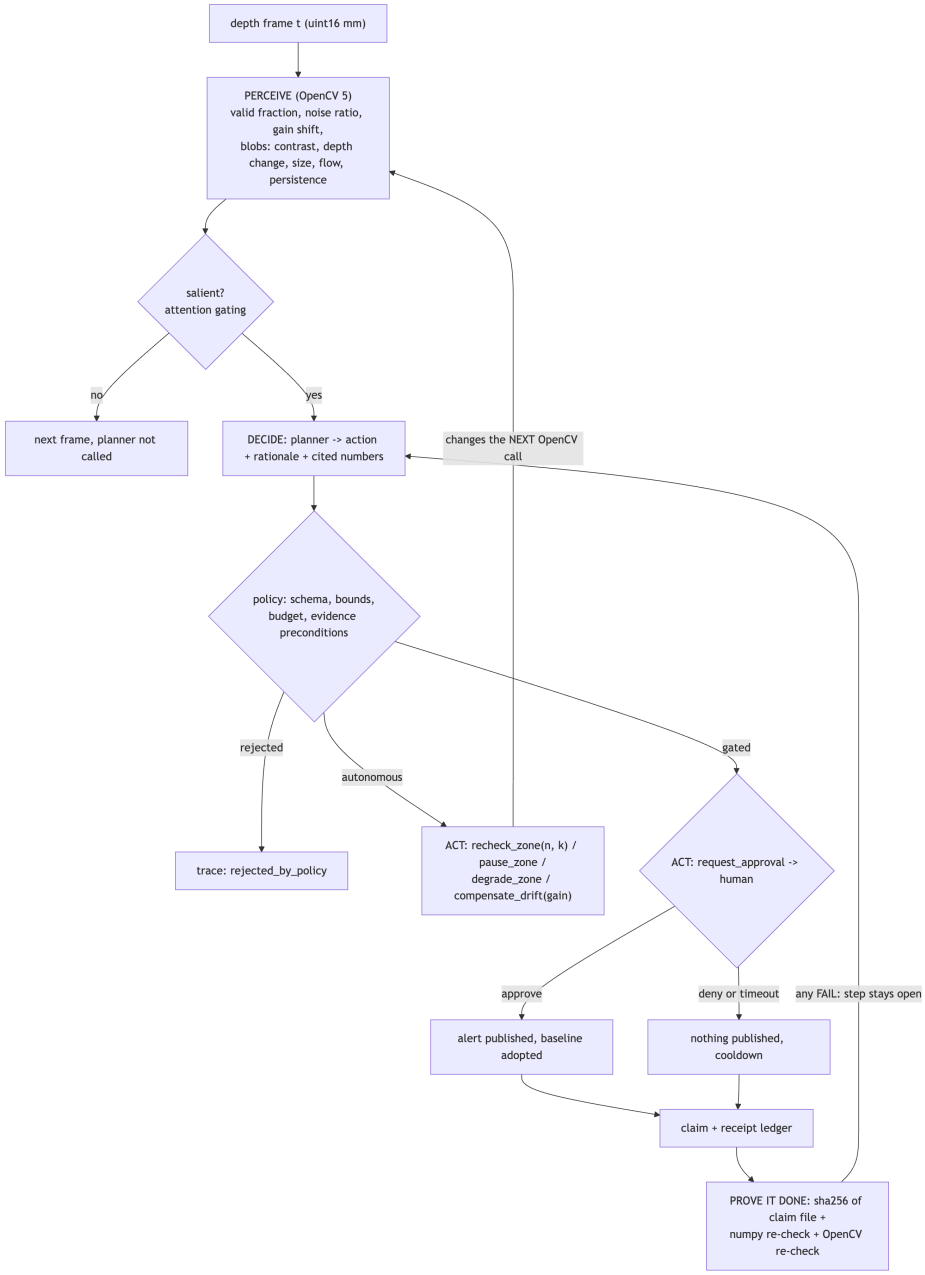


Figure 2. Perception, decision, action, and proof. The loop closes in two places: an autonomous action changes the next OpenCV call, and a failed proof reopens the step.

4.1 Tools, autonomy, and who may do what

The planner is an interface, `plan(view) -> Action`, and sees the same JSON evidence a human would, never pixels. Tools:

Tool	Tier	Effect
`continue`	autonomous	keep watching
`recheck_zone(n, k)`	autonomous	integrate `n` frames (3 to 40) and re-run OpenCV with threshold `k`; at most 3 per episode
`pause_zone(frames)`	autonomous	a person-sized, nearer-than-scene blob is present: do not judge scene changes meanwhile
`degrade_zone(frames)`	autonomous	noise burst or partial depth loss: wait, then escalate if it persists
`compensate_drift(gain)`	autonomous	correct sensor gain drift instead of reporting a phantom change; capped at 20% total
`request_approval(kind)`	**gated**	a human must approve before anything is published

Tiers are enforced by the **executor**, not the planner. `RulePlanner` is deterministic and is what evaluation runs. `LLMPlanner` is an interface plus tests with a fake model: it receives the same JSON view, must answer in the same schema, and anything outside policy is rejected and replaced by the rule planner's choice, with the rejection recorded in the trace. **No language model is called in any result in this report**; the planner here is rule-based. An LLM can be added without widening what the system is allowed to do.

Evidence guardrails. An object claim is refused unless the *current* observation shows a confident, persistent, non-occupant blob in a healthy zone. A sensor claim is refused if coverage and gain look healthy. A planner cannot assert what the pixels do not show.

Attention gating. The planner is called only when an observation is salient (quiet scenes: about 1% of zone-frames), which keeps a future LLM planner affordable.

Human gate. `decide()` returns approve, deny, or nothing; nothing is deny. After a decision a cooldown stops a persistent condition from nagging every frame. Approval adopts the new scene as baseline.

Observability. Every planner call writes a trace record (frame, zone, observation summary, action, rationale, cited numbers, outcome, planner time) to `trace.jsonl`, with counters for calls, rechecks, pauses, rejections and timeouts.

4.2 A worked trace

Scenario `subtle_item_removed`, seed 0: a low item (about 25 mm of relief) is taken at frame 95. One frame sits at about 3.5 sigma, which is not enough to bother a person.

Frame	Observation (OpenCV 5)	Action	Why
95, 96	contrast 3.71 / 3.46, change 27 / 25 mm, persisted 1 / 2 frames	`continue`	ambiguous, not yet persistent
97	contrast 3.34, change 24 mm, persisted 3 frames	`recheck_zone(n=7, k=2.0)`	contrast is between 2.0 and 4.5: integrate 7 frames and look again
104	integrated 7 frames: contrast 5.23, change 24 mm, persisted 10 frames	`request_approval(object_removed)`	clear change held for 10 frames

The threshold-and-alert baseline produces no alert on this scene. The evidence changed what the agent did.

4.3 Receipts

Each claim ("a 892 cm2 region became farther by 24 mm and stayed so for 10 frames") is a done step in a ledger with three receipts through **different instruments**:

1. `file:` SHA-256 of the claim JSON.
2. `cmd:` a numpy-only re-check (never imports `cv2`; a test enforces it).
3. `cmd:` an OpenCV frame-statistics re-check (`cv2.mean`, `cv2.meanStdDev`, `cv2.countNonZero`).

The vendored prove-it-done verifier (pinned to commit `d897534`) runs with `--min-instruments 2`. A step with any failing receipt stays open. A claim pins its input by content digest (SHA-256 over the depth bytes), so anyone can re-run the checks:

```
guetteur verify-ledger runs/<run>/receipts.md
```

Tolerance, stated plainly. A claim passes if the re-measured value is within max(8 mm, 25% of the claim) (coverage +/-0.08, gain +/-0.01). Small inflations pass by design. **Independence is partial:** the two checks share the raw input and the claimed region and window, but not each other's estimator or the live pipeline's thresholds. The receipts prove a claim is *consistent with the pinned input*, not that the input was the right one.

Live operation (planned, not implemented). The raw depth stream is not retained, so a live claim would re-run only against reduced evidence or a short snippet kept after a human approved an escalation. A live claim without a snippet is to be labelled "evidence reduced; not re-derivable from raw" and never presented as fully re-checkable.

5. OpenCV 5 pipeline

Pinned: `opencv-python-headless==5.0.0.93` (`cv2.__version__ == "5.0.0"`), `numpy==2.5.4`. The headless wheel has a manylinux aarch64 build, so the same pin runs on Apple Silicon and on Lambda arm64.

Stage	OpenCV 5 call	Purpose
1. holes	`cv2.inpaint(TELEA)` on 16-bit depth	fill invalid stereo pixels without inventing data when coverage is gone
2. denoise	`cv2.medianBlur(5)` on 16-bit	remove speckle; its effect is *measured at calibration*, not assumed
3. change	signed difference to a temporal-median baseline, in units of a fitted range-dependent noise model	per-pixel significance
4. clean-up	`cv2.morphologyEx` OPEN then CLOSE, elliptical kernel	remove specks, merge fragments
5. blobs	`cv2.connectedComponentsWithStats`	regions; physical size via the pinhole model
6. motion	`cv2.calcOpticalFlowFarneback` on 8-bit depth renderings	moving versus static; sway of a person
evidence	`cv2.resize(INTER_AREA)` to 6x8	the only spatial evidence that may leave the device
receipt #2	`cv2.mean`, `cv2.meanStdDev`, `cv2.countNonZero`	independent per-frame region statistics

Per-call timings on a 168x144 zone crop (Apple M4 Pro, CPU only, `guetteur bench`):

Call	median ms	p95 ms
`inpaint_TELEA_16U`	1.500	1.650
`medianBlur_5_16U`	0.067	0.076
`morphologyEx_open_3x3`	0.007	0.008
`connectedComponentsWithStats`	0.150	0.208
`calcOpticalFlowFarneback`	1.319	1.433
`resize_INTER_AREA_6x8`	0.004	0.004

Whole loop, both zones, per frame (perception + planner + executor): median **8.6 ms**, p95 11.6 ms, p99 12.4 ms, against a 200 ms budget at 5 fps. We have not benchmarked against OpenCV 4 and make no OpenCV 4 comparison claim. Fresh Lambda request timings are reported in section 6; these are individual samples, not a cold-start benchmark.

6. AWS deployment

One SAM stack (`INFRA/template.yaml`), serverless, nothing idle (no NAT gateway, VPC, EC2, load balancer or provisioned capacity):

- **Lambda**, python3.12, **arm64 (Graviton)**, 1769 MB (one full vCPU), 25 s timeout. The arm64 layer carries `opencv-python-headless` and `numpy`; the layer is 136 MB unzipped, under the 250 MB limit.
- **API Gateway HTTP API**, HTTPS, JSON access logs.
- **CloudWatch** logs, alarms on Lambda errors and throttles, X-Ray tracing; **SNS** for alarm mail; an **AWS Budget** with alerts at 50% and 80% of a monthly cap chosen by the account owner.

Guardrails, checked statically by `deploy/check_guardrails.py` and read back after deployment by the deploy script:

Guardrail	Setting
Concurrency cap	no reservation in this deployment: the account quota could not support one. Template parameter 0 omits the reservation; it does not set Lambda concurrency to zero. API throttling remains enabled
API throttling	5 requests/s, burst 10 on every route
Log retention	7 days on both log groups
No public S3	the stack creates no bucket; the phase-2 bucket is private with a full public-access block
Public surface	read-only routes; state-changing routes (`/api/evidence`, `/api/approvals*`) are not deployed and fail shut without a server-side key

Credential-free deploy. The deploy is run by the account owner inside AWS CloudShell under the console session: no access keys are created or stored. `deploy/make-artifact.sh` builds a tarball containing the prebuilt arm64 layer and function, so CloudShell needs neither Docker nor Python 3.12; `deploy/cloudshell-deploy.sh` checks the artifact's checksums and that the layer is an aarch64 build, runs `sam deploy` non-interactively, then smoke-tests `/healthz` (must report `opencv 5.0.0` on `aarch64`) and `/api/run`.

Validation done locally: `sam validate --lint`, `cfn-lint`, `sam build`, `bash -n` and `shellcheck` on the scripts, a stubbed dry run of the whole script, and an import-and-run of the built layer and function on a **linux/arm64 AWS Lambda base image** (healthz reports `opencv 5.0.0`, `aarch64`; scenario passes with receipts proven). Those were pre-deployment checks. The endpoint was subsequently deployed; fresh runtime checks are recorded below.

Endpoint: https://nm4p814bp0.execute-api.us-west-2.amazonaws.com/

Fresh deployed checks (2026-10-14, seed 0, synthetic):

Check	Observed result	Server total
Subtle removal, rule planner	one recheck, one true alert, no miss	5375.9 ms
Same input, naive planner	no recheck, one miss	5415.0 ms
Shelf removal, gate timeout	one request, one timeout, zero published alerts	4660.6 ms
Lens obstructed	one true sensor-fault alert	2539.8 ms
Subthreshold removal	documented miss retained	4767.5 ms
Double the subtle-removal claim	verifier reports `proven: false`	not benchmarked

These are single request samples reported by the deployed handler, not latency percentiles or an isolated cold-start measurement. JSON outputs: `docs/submission/`. Cold-start latency remains unmeasured.

7. Evaluation

7.1 Method

Eleven synthetic scenarios, three seeds each, 33 runs per planner. Scenes are 320x240 uint16 depth with ground truth for events. Scoring is event-level on approval *requests*. A scenario succeeds when every expected alert is raised in its time window, no other alert appears, and the required behaviours

(recheck, pause, drift compensation) are present. Three planners, same evidence: the **rule agent**; **naive + guardrails** (threshold-and-alert with the executor's guardrails); **naive** (threshold-and-alert, no agent, no guardrails).

Scenarios: control (`empty`); `shelf_item_removed`; `shelf_item_added`; `subtle_item_removed` (ambiguous, about 3.5 sigma); `person_walkthrough`; `person_takes_item`; `noise_burst`; `slow_drift`; `lens_obstructed`; `partial_washout`; and `subthreshold_item_removed` (a 10 mm change, **expected to be missed**).

7.2 Results (as measured; re-run confirmed)

Planner	scenarios succeeded (all 11 / excl. documented miss)	true alerts	false alerts	misses	precision	recall (excl. documented miss)	median latency
Guetteur agent (rule planner)	30/33 / 30/30	18	0	3	1.00	0.86 (1.00)	9 frames (1.8 s at 5 fps)
naive + executor guardrails	12/33 / 12/30	9	9	12	0.50	0.43 (0.50)	18 frames (3.6 s)
naive, no agent, no guardrails	12/33 / 12/30	9	27	12	0.25	0.43 (0.50)	9 frames (1.8 s)

- **Receipts:** 21/21 honest ledgers fully proven by the verifier (`--min-instruments 2`).
- **Tamper tests:** 24/24 doctored ledgers rejected. Doctoring: claim inflated 2x; pointed at the wrong pixels (objects) or wrong time (sensor, drift); re-pointed at another recording; the attacker recomputes the `file:` hash.
- **Planner use:** 1693 planner calls on 4605 frames across all scenarios (0.37 per frame, including long eventful stretches); the `empty` control triggers 9 calls in 375 frames.
- **Loop runtime:** median 8.6 ms per frame (both zones), p95 11.6 ms.

Re-run. On 2026-10-12 the full evaluation (`guetteur eval`, 3 seeds) was run again for this report. The scored output was **byte-identical** to the earlier log (`logs/eval.txt` versus `logs/eval-rerun.txt`): 30/33, precision 1.00, recall 0.86 (1.00 excluding the documented miss), tamper 24/24, ledgers 21/21, naive 12/33 at precision 0.25 (0.50 with guardrails). Only wall-clock timing fields in `eval/results.json` differed between the two runs (machine load), so the committed results file keeps the original timings; the loop-runtime figures above come from `guetteur bench` (`eval/runtime.json`).

7.3 Per scenario (seed 0; all seeds in `eval/results.json`)

Scenario	agent	naive + guardrails	naive (raw)	agent alerts (frame, kind)
`empty`	ok	ok	ok	none
`shelf_item_removed`	ok	ok	ok	104: object_removed
`shelf_item_added`	ok	ok	ok	104: object_added
`subtle_item_removed`	ok	missed, never rechecks	missed, never rechecks	104: object_removed
`person_walkthrough`	ok	never pauses	3 false alerts, never pauses	none
`person_takes_item`	ok	never pauses	3 false alerts, never pauses	160: object_removed
`noise_burst`	ok	ok	ok	none
`slow_drift`	ok	3 false alerts, never compensates	3 false alerts, never compensates	none
`lens_obstructed`	ok	missed	missed	107: sensor_fault
`partial_washout`	ok	missed	missed	134: sensor_degraded
`subthreshold_item_removed`	missed (documented)	missed (documented)	missed (documented)	none

Behaviours this shows: ambiguous evidence triggers a re-look instead of an alert or a shrug; presence is not a change (the zone pauses while a person stands in it); sensor trouble is a sensor event, not a scene event; with the human gate set to time out, nothing is published.

7.4 Live-camera results

Three depth-only takes were recorded on 2026-10-14. Each contains 300 native 640x400 frames over approximately 60 seconds. Archive hashes, frame counts, timestamps and clean shutdown were checked. The operator confirmed the walkthrough and item removal. Raw depth stays private; the sanitized integrity manifest is `docs/submission/capture-integrity.json`.

Condition	clips	frames	archive hash	shutdown	real-scene detection result
empty baseline	1	300	verified	clean	not measured
person walkthrough	1	300	verified	clean	not measured
item removed	1	300	verified	clean	not measured

These are capture-integrity results, not accuracy results. Exact event intervals, calibrated geometry/noise and a held-out evaluation split remain open. Native 640x400 frames must not be silently treated as the synthetic 320x240 geometry. No real-world accuracy claim is made.

7.5 What we learned from deployment and capture

- **Camera shutdown is part of correctness.** A capture could reach its frame target and still crash the SDK during teardown. Physical A/B checks isolated the stop order; stopping CAM_C before

CAM_B and draining queues produced clean handoffs. Recovery was checked independently of archive integrity.

- **Deployment validation needs the service API.** The initial HTTP API deployment failed because its access-log format omitted `$context.requestId`. Correcting the format allowed deployment; a passing local template check had not established deployability.
- **Cloud quotas change the guardrail set.** The account could not support the requested concurrency reservation. The deployment records that omission explicitly instead of claiming a cap that is absent.
- **Evidence categories must stay separate.** Recorded frames establish what was captured; operator labels establish what was intended; calibrated replay establishes detector performance. None substitutes for the next.
- **Verification must be challenged.** The tamper suite exposed a flawed test: moving a whole-frame sensor fault to another region leaves it true. The revised wrong-time test and the original 23/24 result are both disclosed.

8. Failure cases

1. **Sub-noise changes are invisible.** `subthreshold_item_removed` (10 mm, about one filtered sigma) is missed by every planner. We report it as a miss, not a pass. Periodic integrated patrol checks would lift the floor; not built.
2. **The tamper test found a hole in its own first definition.** The first run rejected 23/24. The miss was `lens_obstructed` with the *wrong place* tamper: the fault covers the whole frame, so relocating the claimed region leaves the claim true. The tamper is therefore defined as *wrong time* for sensor and drift claims and *wrong pixels* for object claims; the numbers above use the corrected definition.
3. **Small inflations pass by design** (tolerance max(8 mm, 25%)). The receipts catch doctoring beyond the tolerance, not any inflation.
4. **A person partly outside a zone** can momentarily read as a small change; the persistence rule absorbs it, at the cost of latency.
5. **Baseline weaknesses we do not hide:** the naive planners miss `lens_obstructed` and `partial_washout` and raise false alerts on people and on drift; these are intended contrasts, but the baseline is a simple one, not a tuned commercial system.

9. Limitations

- **Synthetic only.** Thresholds were chosen while developing against the same scene family they are scored on, so the figures are optimistic for real data. The noise is Gaussian and range-scaled in the generator; real stereo has structured errors (edges, reflective and dark surfaces, infrared interference).
- **Depth cannot say what an object is.** Claims say "a region became farther by 24 mm", not "a bottle was removed".
- **Latency is dominated by the persistence rule** (10 frames = 2 s at 5 fps); a real deployment would use a longer hold to avoid nuisance alerts.

- **Receipt independence is partial** (shared input, shared claimed region); a wrongly pinned input cannot be caught, only identified by its digest.
- **The planner is rule-based.** The LLM slot is an interface and is not wired to a model. The loop still satisfies "image or video results influence a subsequent plan, tool call, action, or request for human approval", and the policy layer is what bounds any future model.
- **Real-time product integration remains open:** the opt-in native depth recorder is verified, but calibrated detector replay and live edge-to-cloud integration are not complete.
- **AWS cold-start performance is unmeasured.** Individual synthetic request timings are recorded above. Authenticated cloud approvals remain code plus local tests only.
- **Single-camera, two fixed zones** in the prototype; no tracking of individuals, by design.

10. Responsible use and disclosure

Final local test run (2026-10-15): `uv run pytest -q` — **94 passed in 99.08 seconds**. Demo narration uses Kokoro-82M, `am_michael` (Apache-2.0); it is synthetic speech, not a cloned voice. Diagram correction: the original figure said "concurrency cap"; this deployment has no reserved-concurrency reservation, as explained in section 6. The integrated live edge path is design/local code, not a deployed camera-to-cloud product.

- Depth-only by construction: no faces, no images, no video stored or sent. Intended for spaces whose owner controls the camera and where people present know it is there; for homes and care settings only with the informed consent of everyone monitored. Not for workplace productivity monitoring, tracking individuals, or covert use. The system cannot say who someone is and is not designed to.
- Human control: alerts require a person's approval; silence means no alert; claims carry their evidence basis; failures and limits are published.
- Truthful labelling: synthetic data is marked synthetic everywhere; the LLM planner is described as an interface until a model is actually wired.
- **AI assistance disclosure.** The entry was built with AI coding assistance. The agent in the product is deterministic code with an optional LLM slot; we do not claim that using an assistant to write the entry is itself the agentic behaviour being judged.
- Public endpoint security: scenario names and small JSON only, no uploads, no visitor-chosen paths, body-size limit, API throttling, no reserved-concurrency reservation in this account, no secrets in the repository.

11. Reproducing this

```
uv sync                                    # pinned: opencv-python-headless==5.0.0.93, numpy=
uv run pytest                              # includes the real verifier and tamper tests
uv run guetteur demo                       # narrated demo, no camera
uv run guetteur eval                       # all scenarios x 3 seeds x 3 planners -> eval/resu
uv run guetteur bench                      # per-frame runtime on this machine -> eval/runtime
uv run guetteur serve                      # the judge dashboard + API at http://127.0.0.1:808
uv run guetteur verify-ledger runs/<run>/receipts.md
```

Environment of the numbers above: Apple M4 Pro, Python 3.12.15, OpenCV 5.0.0, NumPy 2.5.4, CPU only. Logs: `logs/eval.txt`, `logs/eval-rerun.txt`, `logs/bench.txt`, `logs/demo.txt`, `logs/pytest-last.txt`.

12. Open work and scope limits

Item	Blocked on
Real-scene accuracy (section 7.4)	exact interval labels, calibration and held-out replay
Cold-start benchmark	isolated, repeated measurement; endpoint and smoke timings already verified
Authenticated edge/cloud approval workflow	phase-2 integration and deployed negative tests
Optional: LLM planner wired behind the same policy	decision on model access

Appendix A. Diagram sources

The Mermaid sources are in `docs/report/diagrams/architecture.mmd` and `docs/report/diagrams/agent-loop.mmd`; the PNGs were rendered from them with `mmdc`.